Chetan Aditya Lakka

AI Engineer | Agent Development | Workflow Automation

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Profile

AI Engineer focused on end-to-end agent development, workflow orchestration and system design. Built local, production-style agent systems using Python, FastAPI, LangGraph, PostgreSQL, pgvector and RAG, with human approvals and automated evaluation. Professional experience in computer-use agent evaluation, tool-use reliability and structured failure analysis. Hands-on work in applied ML, data visualization, API integration and reproducible evaluation across text and vision models.

Skills

- **AI / LLM Systems** — Agentic AI, Large Language Models, LangGraph, LangChain, RAG, LLM Agents, Prompt Engineering, OpenAI API, LLM Evaluation, RLHF, SFT, AI Reliability, Multimodal AI, Hugging Face
- **Programming / ML** — Python, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV
- **Backend / Data** — FastAPI, REST APIs, PostgreSQL, pgvector, SQLAlchemy, Pydantic, Alembic, SQL, JSON
- **Testing / Engineering** — Docker, Docker Compose, Pytest, Playwright, Git, API Integration, Model Deployment
- **Explainability / Evaluation** — SHAP, Integrated Gradients, Captum, Model Interpretability, Failure Analysis, Evaluation Design

Projects

ResolveAI - Agentic AI Reliability & IT Access Automation, Python | FastAPI | LangGraph | PostgreSQL | pgvector | RAG | Next.js | Docker

- Designed and built a local, production-style agent automation system using Python, FastAPI, LangGraph and PostgreSQL/pgvector, translating repository-access requests into policy-grounded workflows with simulated tools and verified completion.
- Orchestrated durable LangGraph workflows combining RAG-based policy retrieval, deterministic authorization, resumable human-in-the-loop approvals, escalation and auditable tool execution.
- Implemented idempotent operations, least-privilege rules, typed tool contracts, action-boundary authorization checks and independent read-after-write verification to control execution failures.
- Engineered an automated evaluation framework to assess policy evidence, resource grounding, tool scope, approval ordering, verification, escalation and end-to-end workflow correctness.
- Release validation reports 12 Chromium E2E tests and 37 deterministic evaluation scenarios covering workflow correctness and recovery; implemented PostgreSQL-backed state persistence and resumable execution.

Reverse Attribution - ML Failure Analysis & Explainability, Python | PyTorch | Captum | BERT | RoBERTa | ResNet | XAI

- Developed Reverse Attribution, a supervised post-hoc failure-analysis framework that identifies counter-evidence by analyzing features with negative attribution toward the known true class.
- Designed Counterfactual Pull and Attribution-Flip (A-Flip) metrics to measure true-class logit recovery under feature interventions and disagreement between competing class explanations.
- Built a reproducible frozen-checkpoint evaluation pipeline across NLP and vision models, generating 232 corrected per-example analyses with deterministic sampling, provenance tracking, visualization, and 123 passing tests.
- Audited and repaired the research implementation to align equations, attribution logic, experimental outputs, figures, and manuscript claims while preserving historical model checkpoints.

CereBro - AI-Enabled Cognitive Screening Application, Android | Python | LangChain | REST APIs | On-Device ML | Cloud Integration

- Built an AI-enabled Android application supporting screening workflows for autism, dyslexia, and dementia using on-device machine learning.
- Integrated LLM-based analysis using Python and LangChain for automated response interpretation and feedback generation.
- Designed REST API and cloud integration workflows connecting mobile inference, AI services, and application logic.

Professional Experience

Turing | Contractor

Computer Use Agent Evaluation & Reliability | Team Lead & QA Jan 2026 – Present | Hyderabad (Remote)

- **Completed nearly 1,500** Computer Use Agent evaluation tasks, assessing multi-step planning, tool execution, instruction adherence, factual accuracy and end-to-end task completion.
- Perform RLHF/SFT-style behavioral evaluation to identify hallucinations, grounding errors, planning failures and incorrect tool use.
- Analyze agent execution trajectories to isolate planning, grounding and UI failures; document patterns for regression analysis.
- Develop evaluation rubrics and acceptance criteria to measure agent behavior, task completion and instruction adherence.
- Promoted to Team Lead and QA. Demonstrated persistence, consistent delivery and ownership of complex evaluation workflows.

AI Training Data Specialist - Video Analysis Sep 2025 – Nov 2025 | Hyderabad (Remote)

- Curated multimodal training data for robotics tasks, human actions and interactive video environments.
- Annotated timestamped action and interaction sequences to support temporal grounding and spatial reasoning in vision-language systems.
- Evaluated multimodal outputs for reasoning gaps, inconsistent action sequences and task-specific failure cases.
- Applied structured data-quality checks to improve annotation consistency and support reliable model evaluation.

Education

B.Tech in Electronics & Computer Engineering, Sreenidhi Institute Of Science And Technology 2021 – 2025

Research

Reverse Attribution: Explaining Model Failures via Counter-Evidence, NCNTAIA Conference

CereBro: A Cloud-Integrated Android Application for Cognitive Disorder Screening, NCNTAIA Conference

Certifications

Artificial Intelligence: Knowledge Representation And Reasoning, NPTEL

LLM Trainer Certification, micro1

AWS Academy Machine Learning Foundations, Amazon Web Services (AWS)

AWS Academy Cloud Foundations, Amazon Web Services (AWS)

AI-ML Virtual Internship, AICTE–EduSkills (Supported by Google for Developers)

Android Developer Virtual Internship, AICTE–EduSkills (Supported by Google for Developers)

AI-ML Virtual Internship, AICTE–EduSkills (Supported by AWS Academy)